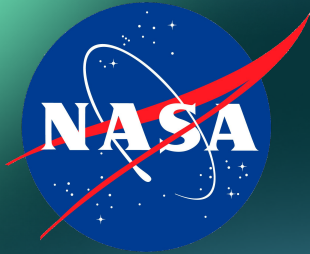


Mitigating Lunar Communication Latency Using an AI-Driven Digital Twin of the Regolith Advanced Surface Systems Operations Robot (RASSOR)



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Introduction

- Space exploration introduces communication delays that make real-time control of robotic systems difficult.
- Planetary missions require intelligent systems capable of operating under latency constraints.
- This project supports FSI in advancing their rover systems by developing and testing a remotely operated RASSOR rover system using RGB-D Camera-based environmental data acquisition and camera-assisted teleoperation, and the foundation for an AI-driven digital twin.



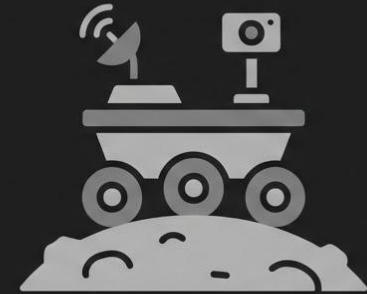
Problem Statement

Current remote rover operations suffer from communication latency, limited environmental awareness, and lack of synchronized feedback systems, reducing operational efficiency and rover safety during remote navigation.

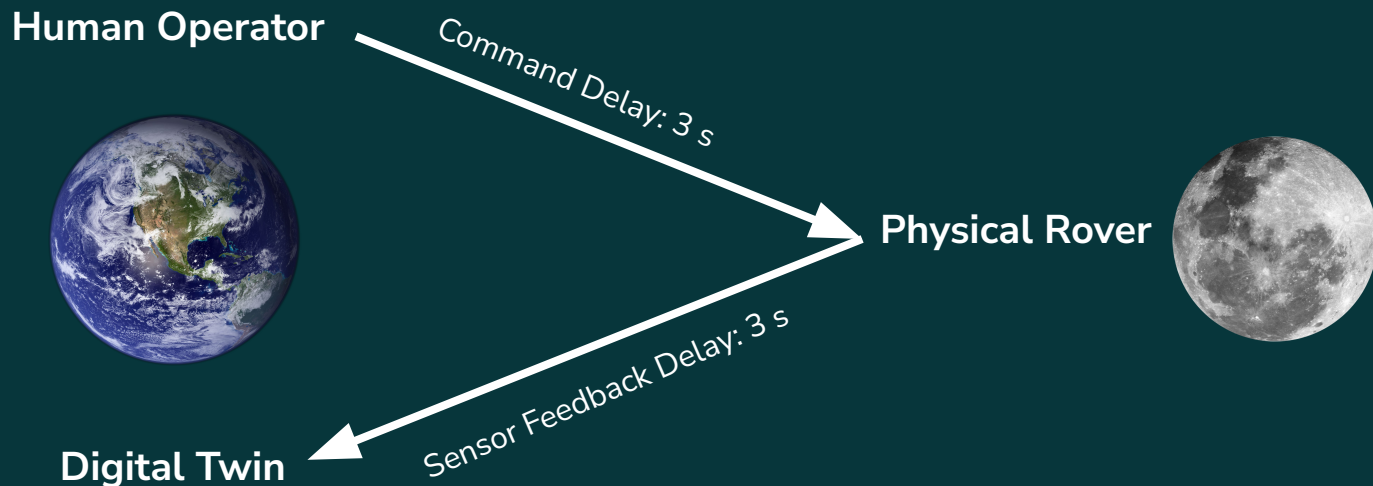


Project Objectives

To tackle in-field communication latency by allowing the digital twin to predict the rover's current state, improving navigation, operator awareness, and overall mission performance.

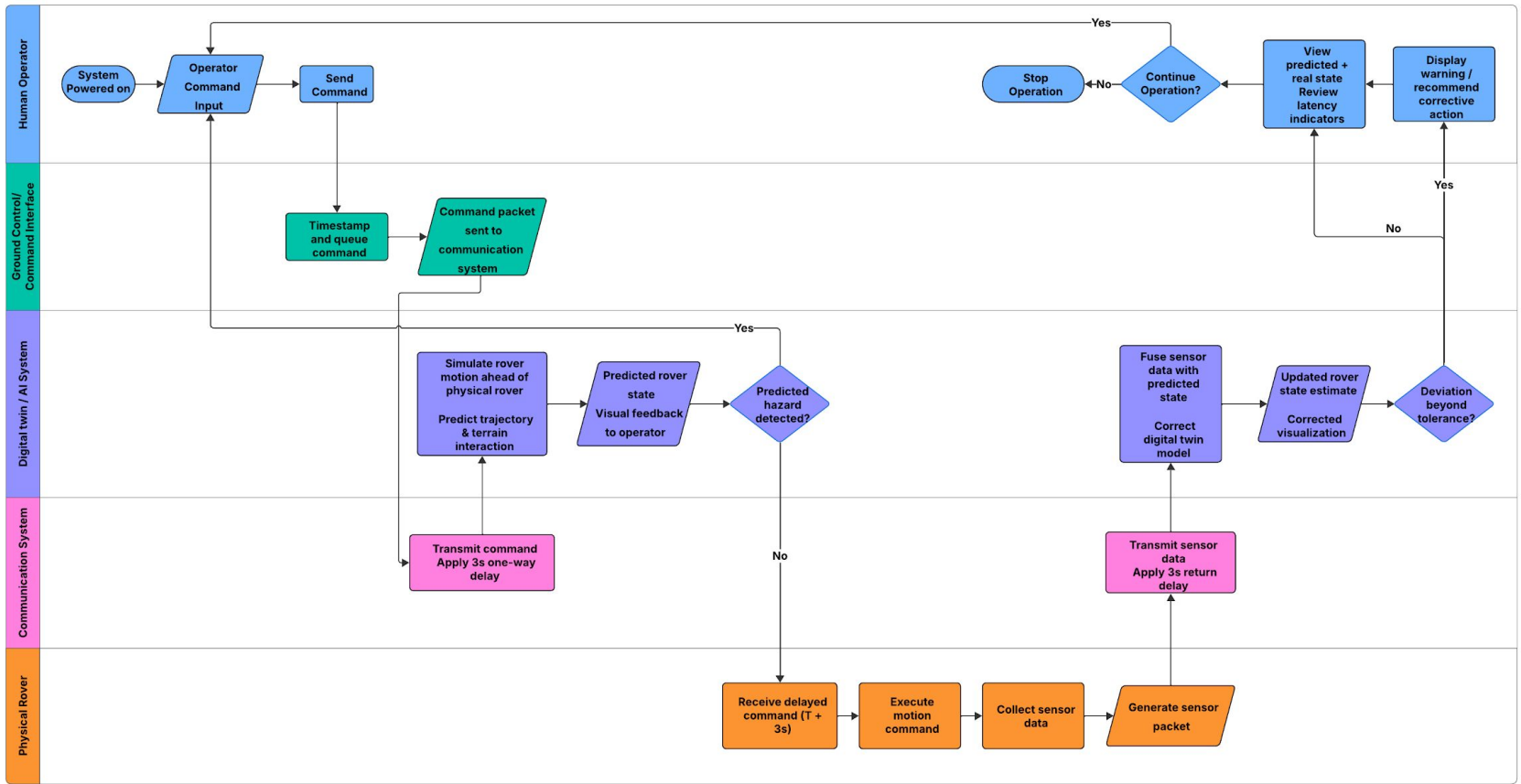


Latency Delay Visualization



Key Insight: Real sensor data received on Earth represents rover state ~3 seconds in the past. The digital twin predicts forward in time to estimate the rover's current state.

Process Map



Methodology

Phase 1:
Requirements
Definition



Phase 2:
Systems
Development



Phase 3:
Digital Twin
Development



Phase 4:
Data
Acquisition



Phase 5:
Analysis &
Evaluation



Phase 6:
Hand-off to
Next Team



Phases 1 & 2

Phase 1: Requirements Definition

- Defined project scope, objectives, and performance metrics:
 - Navigation accuracy
 - Obstacle collisions
 - Task completion time
- Mapped key system components:
 - Operator, rover, communication delay, digital twin
- Identified system constraints:
 - Budget, hardware availability, simulated environment
- Reviewed prior rover projects to guide design

Phase 2: System Development

- Assembling RASSOR rover platform
- Utilized 3D printing for custom components
- Identified and resolved assembly challenges
- Integrated camera system and electronic components
- Used Tinkercad for part design and fit validation

Physical Model

- Custom RASSOR chassis design
- Arduino Mega + RAMPS 1.4 controller
- Stepper motor drive system
- Adjustable speed and torque control



Initial Command Delay

	Speed 1	Speed 2	Speed 3
Command	Average Delay (s)		
Forward	19.7	18	34
Back	33.7	18.3	16.3
Right	25.3	16.7	7.7
Left	26	31.3	24.7
Stop	24.9	13.5	13.5

Feedback:

- Speed ≠ faster response
- Big variability = system issue

Solution:

- Identified software delay
- Removed extra code loop
- Achieved intentional 3s Delay

Phases 3 & 4

Phase 3: Digital Twin Development

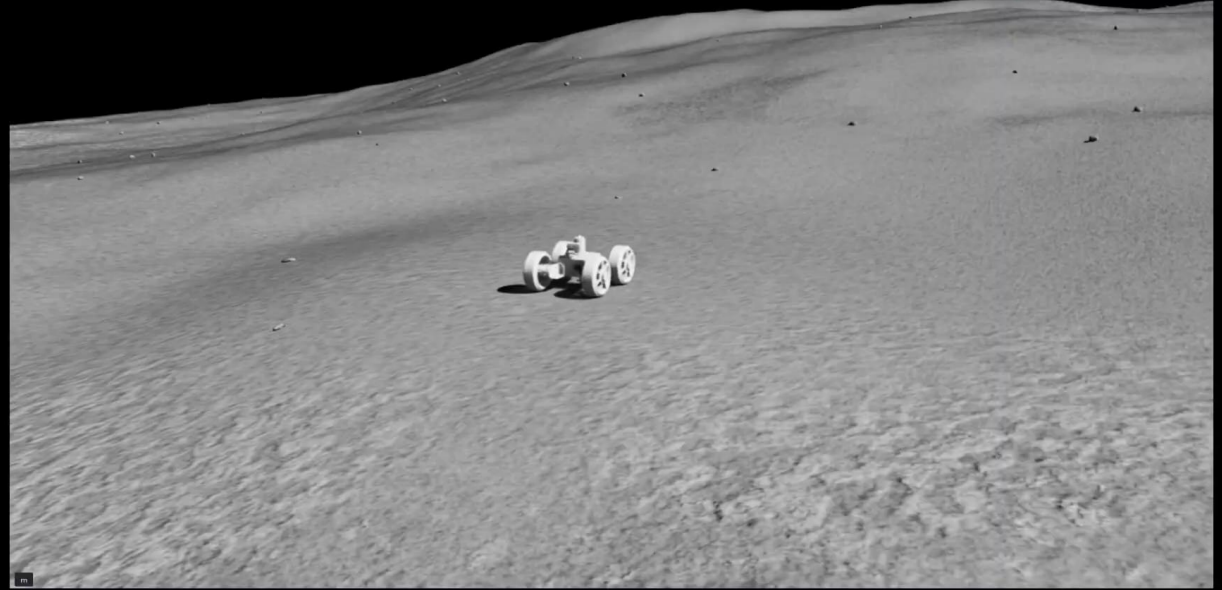
- Researched multiple digital twin platforms on NVIDIA Omniverse
- Evaluated options based on:
 - Compatibility with rover system
 - Ability to simulate motion and environment
 - Ease of integration with sensor data
- Selected Isaac Sim the most suitable digital twin system before full implementation because of its further development geared towards robotic simulation and testing.

Phase 4: Data Acquisition

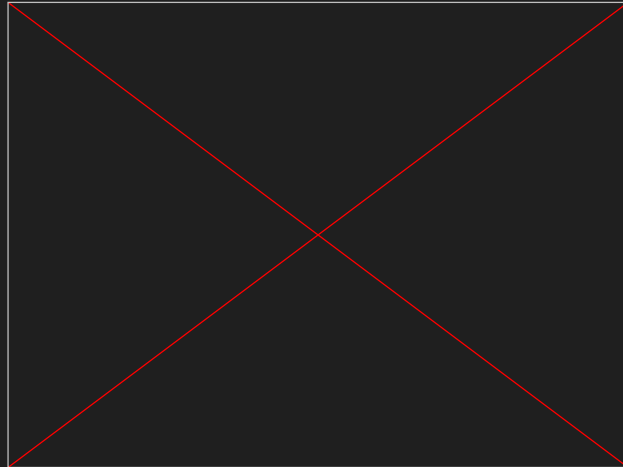
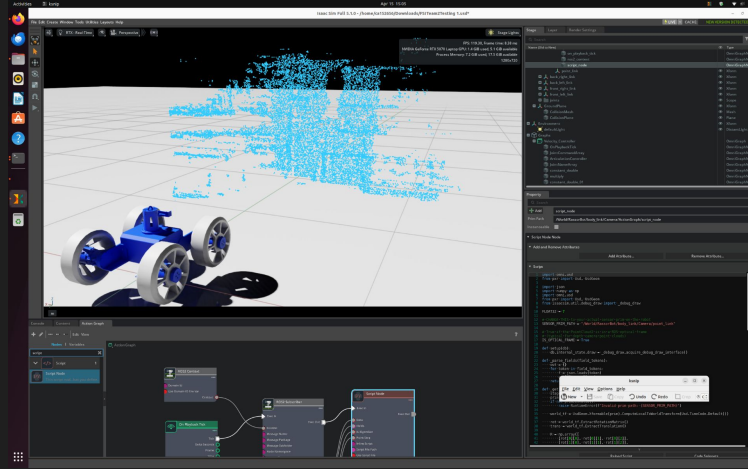
- Collect real-time point cloud data using the OAK-D Lite camera system
 - Enables 3D mapping of the rover's environment
- Capture visual feedback through onboard camera for remote operation
- Recording rover motion data: Position, velocity, and trajectory
- Stream and store sensor data for use in digital twin development
- Rover uses Microsoft VSCode to operate

Digital Model Video

- CAD model imported from OnShape
- Physics-based rover simulation
- Configured joints and colliders
- Foundation for predictive modeling



Point Cloud Data from RGB-D Camera



3D Mapping

Phases 5 & 6

Phase 5: Analysis & Evaluation

- Conduct controlled testing in an obstacle course under simulated communication delay
- Compare system performance across configurations:
 - Manual teleoperation (baseline)
 - Digital twin-assisted operation
- Evaluate key performance metrics:
 - Task completion time
 - Collision frequency
 - Path deviation (navigation accuracy)
- Identify performance gaps and quantify improvements from predictive modeling

Phase 6: Hand-off to Next Team

- Develop comprehensive technical documentation:
 - System architecture
 - Design decisions
 - Testing procedures and results
- Package digital twin framework for:
 - AI integration and autonomous control development
- Provide structured guidelines for:
 - Continued system testing
 - Model refinement and validation
- Ensure knowledge transfer to maintain project continuity across semesters

Results and Findings

Results

- Achieved functional digital mapping using RGB-D camera data
- System upgrades improved performance by switching from Le Potato SBC -> Raspberry Pi
- Re-enabled center camera with higher quality output for improved mapping accuracy

Key Findings

- Battery life is a major constraint (~1 hour runtime vs. ~3 hour recharge)
- High power demand from simultaneous sensing + mobility
- Hardware limitations impacted data throughput and reliability
- Tradeoff observed between data quality and system performance due to power



Final Recommendations

System Improvements

- Implement a dual-battery system to separate camera and motor power loads
- Redesign/extend rover base to accommodate additional hardware

Performance Optimization

- Improve power management strategies to extend operational time
- Balance sensor quality vs. processing capability for efficient mapping

Hardware & Integration

- Continue using higher-performance computing (e.g., Raspberry Pi or better alternatives)
- Optimize data transfer and processing pipelines

Future Implementation

- Maintain full sensor suite (including center camera) for better environmental accuracy
- Support future integration of AI-based navigation and obstacle avoidance

Project Deliverables

A completed, drivable
rover prototype, outfitted
with working sensors for
data collection



Functioning digital twin
system designed for
prediction of rover's
motion



Setup framework for next
team towards AI assisted
Rover control



Ideas on Future Work

- Transition project into Phase 2 development with a new team focused on AI integration
- Leverage the established digital twin framework to support AI-driven autonomy
- Conduct incremental field testing in controlled sandy environments before full deployment
- Validate rover performance in regolith-like conditions (e.g., Exolith Lab testing)
- Expand from digital mapping → real-time obstacle detection and avoidance
- Enhance teleoperation with semi-autonomous capabilities under latency constraints